

MATH1061 26s1 Final Practice

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W1 Sets, Numbers, and Complex Numbers

Q1.

Let $A = \{x \in \mathbb{R} : x^2 < 4\}$ and $B = [0, 3)$. What is $A \cap B$?

- (A) $[0, 2)$
- (B) $(0, 2]$
- (C) $(-2, 3)$
- (D) $[0, 3)$

Q2. **Pitfall**

Which statement is logically equivalent to $A \subseteq B$?

- (A) If $x \in B$, then $x \in A$.
- (B) If $x \in A$, then $x \in B$.
- (C) $A \cap B = \emptyset$.
- (D) $A \cup B = A \cap B$.

Q3.

For $z = -1 + \sqrt{3}i$, the principal argument is

- (A) $\pi/3$
- (B) $-\pi/3$
- (C) $2\pi/3$
- (D) $-2\pi/3$

Q4.

Which is equal to $1/(3 - 4i)$?

- (A) $(3 - 4i)/25$
- (B) $3 + 4i$
- (C) $(4 + 3i)/25$
- (D) $(3 + 4i)/25$

Q5. **Formula**

All solutions of $z^3 = 8e^{i\pi}$ have modulus

- (A) 2
- (B) 8
- (C) $\sqrt[3]{\pi}$
- (D) 1

Q6.

Which set describes quadrants II and IV, excluding axes?

- (A) $\Re(z)\Im(z) > 0$
- (B) $\Re(z)\Im(z) < 0$
- (C) $\Im(z) > 0$
- (D) $\Re(z) < 0$

Q7.

If $p(z)$ has real coefficients and $1 + i$ is a root, then another root must be

- (A) $-1 + i$
- (B) $-1 - i$
- (C) $1 - i$
- (D) $i - 1$

**Q8. Novel**

Let $z = 2 - 2i$.

- (a) Write z in polar exponential form.
- (b) Compute z^4 in rectangular form.
- (c) Explain why using $\theta = \pi/4$ would give the wrong quadrant.

Q9. Strategy

Solve $w^4 = 16i$.

- (a) Write $16i$ in polar form.
- (b) Find all four roots in exponential form.
- (c) List their arguments in $[0, 2\pi)$.

Q10. Pitfall

A student claims $(A \cup B)^c = A^c \cup B^c$.

- (a) Give a concrete counterexample with subsets of $\{1, 2, 3\}$.
- (b) State the correct De Morgan law.
- (c) Briefly explain the logical reason for the correction.



W2 Functions and Complex Numbers

Q1.

For $f: \mathbb{Z} \rightarrow \mathbb{Z}$, $f(n) = 2n + 1$, which is true?

- (A) Surjective but not injective.
- (B) Injective but not surjective.
- (C) Bijective.
- (D) Neither.

Q2. Pitfall

Let $g: \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = x^2$. Which is true?

- (A) Injective not surjective.
- (B) Surjective not injective.
- (C) Neither injective nor surjective.
- (D) Bijective.

Q3.

The natural domain of $\ln(4 - x^2)$ is

- (A) $[-2, 2]$
- (B) $(-\infty, -2) \cup (2, \infty)$
- (C) $\mathbb{R} \setminus \{\pm 2\}$
- (D) $(-2, 2)$

Q4.

If $z = 3e^{i\theta}$, then $|z^5|$ is

- (A) 243
- (B) 15
- (C) 3
- (D) $3e^{5i\theta}$

Q5. Formula

Which restriction makes x^2 invertible?

- (A) Domain $\mathbb{R}$, codomain $[0, \infty)$.
- (B) Domain $[0, \infty)$, codomain $[0, \infty)$.
- (C) Domain $\mathbb{R}$, codomain $\mathbb{R}$.
- (D) Domain $(-\infty, \infty)$, codomain $(0, \infty)$.

Q6.

If f is strictly decreasing on an interval, then f is

- (A) Surjective onto $\mathbb{R}$.
- (B) Even.
- (C) Injective on that interval.
- (D) Periodic.

Q7.

Which complex number has argument $-\pi/2$?

- (A) $2i$
- (B) -2
- (C) 2
- (D) $-2i$

**Q8. Strategy**

Consider $F(x) = \frac{\ln(1+3x) - \ln(1+x)}{x}$.

- (a) Find the natural domain excluding any removable point.
- (b) Compute $\lim_{x \rightarrow 0} F(x)$.
- (c) Define a continuous extension at $x = 0$.

Q9. Novel

Let $h : [-1, \infty) \rightarrow [0, \infty)$, $h(x) = \sqrt{x+1}$.

- (a) Prove h is injective.
- (b) Prove h is surjective onto the stated codomain.
- (c) Find $h^{-1}(y)$ and its domain.

Q10. Pitfall

A student writes that $z^3 = 8$ has only solution $z = 2$.

- (a) Find all complex solutions.
- (b) Explain why the real-number intuition misses two roots.
- (c) Sketch the root arguments using a tiny unit-circle diagram if desired.



W3 Limits and Vectors

Q1.

$$\lim_{x \rightarrow 0} \frac{\sin(4x)}{x} \text{ equals}$$

- (A) 1
- (B) 0
- (C) 4
- (D) Does not exist

Q2. **Formula**

$$\lim_{x \rightarrow 0} \frac{1 - \cos(3x)}{x^2} \text{ equals}$$

- (A) $3/2$
- (B) 3
- (C) 0
- (D) $9/2$

Q3.

If $\mathbf{a} \cdot \mathbf{b} = 0$ and both vectors are nonzero, the vectors are

- (A) Perpendicular.
- (B) Parallel.
- (C) Equal.
- (D) Opposite.

Q4.

$$(1, 2, 0) \times (0, 1, 3) \text{ equals}$$

- (A) $(3, -6, 1)$
- (B) $(6, -3, 1)$
- (C) $(1, -3, 6)$
- (D) $(6, 3, -1)$

Q5. **Pitfall**

Which limit form is indeterminate?

- (A) $5/0^+$
- (B) $0/5$
- (C) $0/0$
- (D) e^0

Q6.

The area of the parallelogram spanned by $\mathbf{a}$, $\mathbf{b}$ is

- (A) $\mathbf{a} \cdot \mathbf{b}$
- (B) $\|\mathbf{a} + \mathbf{b}\|$
- (C) $\|\mathbf{a}\|\|\mathbf{b}\|$ always
- (D) $\|\mathbf{a} \times \mathbf{b}\|$

Q7.

$$\text{proj}_{(1,2)}(3, 0) \text{ is}$$

- (A) $(3/5, 6/5)$
- (B) $(3, 0)$
- (C) $(1, 2)$
- (D) $(6/5, 12/5)$

**Q8. Novel**

Use Taylor expansions to evaluate $L = \lim_{x \rightarrow 0} \frac{\sin(2x) - 2x + \frac{4}{3}x^3}{x^5}$.

- (a) Expand $\sin(2x)$ through the x^5 term.
- (b) Identify the first nonzero surviving numerator term.
- (c) Compute L .

Q9. Strategy

Let $\mathbf{a} = (1, -1, 2)$, $\mathbf{b} = (3, 0, 1)$.

- (a) Compute $\mathbf{a} \cdot \mathbf{b}$.
- (b) Compute $\mathbf{a} \times \mathbf{b}$.
- (c) Find the parallelogram area.

Q10. Pitfall

A student uses $\sin(3x) \sim 3x$ and $\sin x \sim x$ to claim $\frac{\sin(3x) - 3\sin x}{x^3} \rightarrow 0$.

- (a) Explain why first-order approximations are insufficient.
- (b) Compute the correct limit.
- (c) State a general lesson for cancellation limits.



W4 Continuity, Differentiation, and Vectors

Q1.

For $h(x) = \begin{cases} x^2, & x < 1 \\ mx + b, & x \geq 1 \end{cases}$, continuity and differentiability at 1 require

- (A) $m = 1, b = 0$
- (B) $m = 2, b = 1$
- (C) $m = 0, b = 1$
- (D) $m = 2, b = -1$

Q2. Formula

If $e^y + y = x + 1$, $y(0) = 0$, then $y'(0) =$

- (A) $1/2$
- (B) 1
- (C) 2
- (D) 0

Q3.

A normal vector to $2x - y + 5z = 7$ is

- (A) $(2, 1, 5)$
- (B) $(2, -1, 5)$
- (C) $(1, 2, 5)$
- (D) $(5, -1, 2)$

Q4.

The line $(x, y, z) = (1, 0, 2) + t(2, 1, -1)$ meets $x + y + z = 5$ when

- (A) $t = 2$
- (B) $t = 0$
- (C) $t = 1$
- (D) never

Q5. Pitfall

If $f'(c) = 0$, which conclusion is justified?

- (A) c must be a local min.
- (B) c must be a local max.
- (C) f is constant.
- (D) c is only a candidate for local extremum.

Q6.

If f'' changes from negative to positive at c , then

- (A) f has a local max at c always.
- (B) concavity changes from down to up.
- (C) $f'(c) = 0$ always.
- (D) $f(c) = 0$.

Q7.

The plane through P with normal $\mathbf{n} = (a, b, c)$ has equation

- (A) $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$
- (B) $(x, y, z) = P + t\mathbf{n}$
- (C) $ax + by + cz = 0$ always
- (D) $\mathbf{n} \times P = 0$

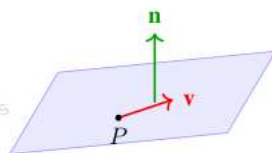
**Q8. Novel**

Let $g(x) = x + \frac{4}{x}$ on $(0, \infty)$.

- Find the critical point.
- Classify it.
- Determine concavity on $(0, \infty)$.

Q9. Strategy

The sketch below shows a line through P with direction $\mathbf{v}$ and a plane with normal $\mathbf{n}$.



- State the condition for the line direction $\mathbf{v}$ to be parallel to the plane.
- State the condition for $\mathbf{v}$ to be perpendicular to the plane.
- If $\mathbf{n} = (2, -1, 1)$, give one nonzero vector parallel to the plane.

Q10. Pitfall

A student classifies $x = 0$ as a local minimum of $f(x) = x^3$ because $f'(0) = 0$.

- Explain the error.
- Use a sign chart or second derivative reasoning to classify correctly.
- Give one function where $f'(0) = 0$ is a genuine local minimum.



W5 Limit Methods and Vector Geometry

Q1.

$\lim_{x \rightarrow \infty} (1 + 3/x)^{2x}$ equals

- (A) e^6
- (B) e^3
- (C) 6
- (D) ∞

Q2. Pitfall

If $\mathbf{A} \times \mathbf{B} = (-1, -1, 2)$, then $(\mathbf{A} + \mathbf{B}) \times (3\mathbf{A} + 5\mathbf{B})$ is

- (A) $(-8, -8, 16)$
- (B) $(-2, -2, 4)$
- (C) $(2, 2, -4)$
- (D) $(0, 0, 0)$

Q3.

Distance from $(1, 2, 3)$ to $x - 2y + 2z = 5$ is

- (A) 2
- (B) $4/3$
- (C) $2/3$
- (D) 0

Q4.

Two nonparallel lines in $\mathbb{R}^3$ that do not intersect are

- (A) parallel
- (B) orthogonal
- (C) identical
- (D) skew

Q5. Formula

$\lim_{x \rightarrow \infty} \frac{2x^2 - 1}{x^2 + 3x}$ is

- (A) 2
- (B) 0
- (C) ∞
- (D) 1

Q6.

A vector parallel to $3x - y + z = 1$ must have dot product with $(3, -1, 1)$ equal to

- (A) 1
- (B) 0
- (C) 3
- (D) -1

Q7.

$\lim_{x \rightarrow 0} \frac{\ln(1+x) - x}{x^2}$ equals

- (A) 1
- (B) 0
- (C) $-1/2$
- (D) $1/2$

**Q8. Strategy**

Let $\ell : (x, y, z) = (1, 2, 0) + t(1, -1, 2)$.

- (a) Find where ℓ intersects $x + y + z = 8$.
- (b) Find the angle relation between the line and the plane normal.
- (c) State when such a line would be parallel to the plane.

Q9. Novel

Evaluate $\lim_{x \rightarrow \infty} (\ln(x+1))^{1/\ln x}$.

- (a) Take logarithms.
- (b) Compute the transformed limit.
- (c) Return to the original limit.

Q10. Pitfall

Given $P = (0, 0, 0)$, $Q = (1, 2, 0)$, $R = (2, -1, 1)$.

- (a) Find vectors spanning the parallelogram PQR .
- (b) Compute its area.
- (c) Explain why dot product would not give the area.



W6 Optimisation and Planes

Q1.

On a closed interval, absolute extrema occur among

- (A) critical points only
- (B) endpoints and critical points
- (C) endpoints only
- (D) inflection points only

Q2. Formula

For $f(x) = x + \frac{1}{x-1}$ on $(1, \infty)$, $f'(x) = 0$ at

- (A) $x = -1$
- (B) $x = 0$
- (C) $x = 2$
- (D) no point

Q3.

The plane through A, B, C has normal

- (A) $(B - A) \cdot (C - A)$
- (B) $A + B + C$
- (C) $(A - B) + (A - C)$
- (D) $(B - A) \times (C - A)$

Q4.

A vector perpendicular to the plane is

- (A) a normal vector
- (B) any point on the plane
- (C) any direction in the plane
- (D) the zero vector only

Q5. Pitfall

If $f''(c) = 0$, then c

- (A) is always a max
- (B) cannot be classified by the second derivative test alone
- (C) is always a min
- (D) is always inflection

Q6.

The distance between parallel planes $2x - y + 2z = 3$ and $2x - y + 2z = 9$ is

- (A) 6
- (B) $6/\sqrt{5}$
- (C) 2
- (D) 3

Q7.

If f' changes from positive to negative at c , then

- (A) local minimum
- (B) no extremum
- (C) vertical asymptote
- (D) local maximum

**Q8. Strategy**

Optimise $A(x) = x(12 - 2x)$ for $0 \leq x \leq 6$.

- (a) Find critical points.
- (b) Compare endpoint and critical values.
- (c) Interpret the maximum.

Q9. Novel

Find the plane through $P = (1, 0, 2)$, $Q = (3, 1, 1)$, $R = (2, -1, 3)$.

- (a) Compute two direction vectors.
- (b) Find a normal vector.
- (c) Write a Cartesian equation.

Q10. Pitfall

Consider $g(x) = \frac{x+1}{x-1}$.

- (a) State the natural domain.
- (b) Find critical points.
- (c) Find asymptotes and concavity on $(1, \infty)$.



W7 Optimisation and Linear Systems

Q1.

A square system has a unique solution iff

- (A) $\det A = 0$
- (B) there is a zero row in A
- (C) $\det A \neq 0$
- (D) the augmented matrix has no pivots

Q2. Formula

If A is 3×3 , $\det A = 4$, then $\det(2A^{-1}) =$

- (A) $1/2$
- (B) 8
- (C) 16
- (D) 2

Q3.

Multiplying one row by 5 multiplies the determinant by

- (A) 5
- (B) $1/5$
- (C) -5
- (D) 0

Q4.

A row replacement operation changes the determinant by

- (A) factor -1
- (B) factor 1
- (C) factor 2
- (D) making it zero

Q5. Pitfall

The system $x + y = 1$, $2x + 2y = 3$ has

- (A) one solution
- (B) infinitely many
- (C) no solution
- (D) two solutions

Q6.

If A^{-1} exists, then $Ax = b$ has

- (A) no solution
- (B) infinitely many
- (C) only the zero solution
- (D) exactly one solution

Q7.

RREF with a pivot in every variable column and no inconsistent row means

- (A) unique solution
- (B) no solution
- (C) infinite solutions
- (D) determinant zero always

**Q8. Strategy**

Solve $x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 2$.

- (a) Write the augmented matrix.
- (b) Use elimination to solve.
- (c) Check the solution in one equation.

Q9. Novel

For what a does the system $x + y = 1$, $2x + ay = 2$ fail to have a unique solution?

- (a) Write the coefficient determinant.
- (b) Find the exceptional a .
- (c) Classify the system at that value.

Q10. Pitfall

A matrix A is reduced to I by one row swap and one row scaling by 3.

- (a) Relate $\det I$ to $\det A$.
- (b) Find $\det A$.
- (c) Explain why row replacement would not affect this calculation.



W8 Taylor Polynomials and Matrices

Q1.

Taylor polynomial of $\ln(1 + 2x)$ to degree 3 is

- (A) $2x - x^2 + \frac{2}{3}x^3$
- (B) $2x + 2x^2 + \frac{8}{3}x^3$
- (C) $x - 2x^2 + \frac{4}{3}x^3$
- (D) $2x - 2x^2 + \frac{8}{3}x^3$

Q2. Formula

The degree 4 Taylor polynomial of e^{-x^2} at 0 is

- (A) $1 - x^2 + \frac{x^4}{2}$
- (B) $1 - x + x^2/2 - x^3/6 + x^4/24$
- (C) $1 + x^2 + x^4/2$
- (D) $-x^2 + x^4/2$

Q3.

If A is 4×4 , $\det(3A) =$

- (A) $3 \det A$
- (B) $81 \det A$
- (C) $12 \det A$
- (D) $\det A/3$

Q4.

$(AB)^T =$

- (A) $A^T B^T$
- (B) AB
- (C) $B^T A^T$
- (D) BA

Q5. Pitfall

A degree 3 Taylor remainder uses which derivative in Lagrange form?

- (A) third derivative
- (B) second derivative
- (C) first derivative
- (D) fourth derivative

Q6.

If A is triangular, $\det A$ is

- (A) product of diagonal entries
- (B) sum of diagonal entries
- (C) product of all entries
- (D) always 1

Q7.

Matrix multiplication AB is defined when

- (A) rows of A equal rows of B
- (B) columns of A equal rows of B
- (C) A, B are square only
- (D) A, B have same size only

**Q8. Strategy**

Let $f(x) = \sqrt{1+x}$.

- (a) Find the Taylor polynomial through degree 3 at 0.
- (b) Determine the sign of the degree 3 Lagrange remainder for $0 < x < 1$.
- (c) Explain why the sign matters for an upper/lower estimate.

Q9. Novel

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

- (a) Compute AB and BA .
- (b) Use the result to explain noncommutativity.
- (c) Compute $\det(AB)$ using determinants.

Q10. Pitfall

A student expands e^{-x^2} as $1 - x^2 + x^4/24 + \dots$.

- (a) Find the first incorrect coefficient.
- (b) Correct the degree 4 polynomial.
- (c) Explain the substitution mistake.



W9 Riemann Integrals and Matrices

Q1.

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^2}{n^3} \text{ equals}$$

- (A) $1/3$
- (B) $1/2$
- (C) 1
- (D) 0

Q2. Formula

$$\text{If } G(x) = \int_0^{x^2} \ln(1+t^2) dt, \text{ then } G'(x) =$$

- (A) $\ln(1+x^4)$
- (B) $2x \ln(1+x^4)$
- (C) $x^2 \ln(1+x^2)$
- (D) $2x \ln(1+x^2)$

Q3.

$$\text{If } f > 0 \text{ and } f' > 0, G(x) = \int_0^x f(t) dt \text{ is}$$

- (A) increasing and concave down
- (B) decreasing
- (C) increasing and concave up
- (D) not determined increasing

Q4.

A homogeneous system $Ax = 0$ is always

- (A) inconsistent
- (B) unique
- (C) nonlinear
- (D) consistent

Q5. Pitfall

Rank-nullity for $A \in \mathbb{R}^{m \times n}$ says

- (A) $\text{rank } A + \text{nullity } A = n$
- (B) $= m$
- (C) $= m + n$
- (D) $= 0$

Q6.

The signed area $\int_a^b f(x) dx$ can be negative when

- (A) $b > a$
- (B) f is mostly below the axis
- (C) f is continuous
- (D) f is increasing

Q7.

If a 3×3 matrix has rank 2, the nullity is

- (A) 2
- (B) 3
- (C) 1
- (D) 0

**Q8. Strategy**

Convert $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} \left(1 + \frac{k}{n}\right)^2$.

- (a) Identify the interval.
- (b) Write the corresponding integral.
- (c) Evaluate it.

Q9. Novel

Let $F(x) = \int_x^{e^x} \sqrt{1+t^2} dt$.

- (a) Differentiate using variable limits.
- (b) Evaluate $F'(0)$.
- (c) Explain the sign of the lower-limit term.

Q10. Pitfall

A student claims $\sum_{k=1}^n k^2/n^3 \cdot n$ is a Riemann sum for $\int_0^1 x^2 dx$.

- (a) Rewrite the expression carefully.
- (b) Determine whether it converges.
- (c) State the missing or extra factor.



W10 Integration Methods and Determinants

Q1.

$$\int_0^1 \frac{x}{1+x^2} dx =$$

- (A) $\ln 2$
- (B) $\frac{1}{2} \ln 2$
- (C) 1
- (D) $\pi/4$

Q2. Pitfall

Which improper integral converges?

- (A) $\int_1^\infty x^{-1} dx$
- (B) $\int_0^1 x^{-2} dx$
- (C) $\int_1^\infty x^{-2} dx$
- (D) $\int_0^\infty 1 dx$

Q3.

If A is transformed to I by one row swap and row scaling by 3, $\det A =$

- (A) 3
- (B) -3
- (C) $1/3$
- (D) $-1/3$

Q4.

The determinant of an upper triangular matrix is

- (A) product diagonal
- (B) sum diagonal
- (C) last diagonal
- (D) zero always

Q5. Formula

$$\int x e^x dx =$$

- (A) $x e^x + C$
- (B) $x e^x - e^x + C$
- (C) $e^x + C$
- (D) $x^2 e^x / 2 + C$

Q6.

If $\det A = 0$, then A is

- (A) invertible
- (B) orthogonal
- (C) singular
- (D) diagonal

Q7.

With $u = \sqrt{t}$, $dt =$

- (A) $du/(2u)$
- (B) $u^2 du$
- (C) du
- (D) $2u du$

**Q8. Strategy**

Evaluate $I = \int_0^{16} \frac{1}{(\sqrt{t}+1)(\sqrt{t}+3)} dt$.

- (a) Use $u = \sqrt{t}$ and change limits.
- (b) Decompose into partial fractions.
- (c) Evaluate the integral.

Q9. Novel

Let $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 0 & 1 \end{pmatrix}$.

- (a) Compute $\det A$.
- (b) Decide if A is invertible.
- (c) State what this means for $Ax = b$.

Q10. Pitfall

A student treats $\int_0^1 x^{-1} dx$ as $[\ln x]_0^1$.

- (a) State why this is invalid as written.
- (b) Evaluate it as an improper integral.
- (c) Compare with $\int_0^1 x^{-1/2} dx$.



W11 Integration and Eigenvalues

Q1.

Eigenvalues of $\begin{pmatrix} 4 & 1 \\ 0 & 2 \end{pmatrix}$ are

- (A) 1 and 0
- (B) 6 and 8
- (C) 4 and 2
- (D) only 4

Q2. Formula

If $Av = \lambda v$, then $A^3v =$

- (A) $3\lambda v$
- (B) λv^3
- (C) $A\lambda^3$
- (D) $\lambda^3 v$

Q3.

Area between $y = 2x$ and $y = x^2$ on $[0, 2]$ is

- (A) $4/3$
- (B) $8/3$
- (C) 2
- (D) $16/3$

Q4.

Washers about the x -axis use

- (A) $2\pi \int \text{radius} \cdot \text{height} dx$
- (B) $\pi \int (R^2 - r^2) dx$
- (C) $\int f'$
- (D) $\sum \lambda_i$

Q5. Pitfall

Which condition is sufficient to guarantee that an $n \times n$ matrix is diagonalizable over $\mathbb{R}$?

- (A) It has a repeated eigenvalue.
- (B) Its determinant is nonzero.
- (C) It has n distinct real eigenvalues.
- (D) Its trace is nonzero.

Q6.

The trace of a matrix equals

- (A) product of eigenvalues
- (B) rank
- (C) determinant
- (D) sum of eigenvalues counting algebraic multiplicity

Q7.

If A has three distinct eigenvalues in $\mathbb{R}^3$, it is

- (A) diagonalizable
- (B) singular
- (C) not diagonalizable
- (D) zero

**Q8. Strategy**

Find eigenvalues and eigenvectors of $A = \begin{pmatrix} 0 & 6 \\ 1 & -1 \end{pmatrix}$.

- (a) Compute the characteristic polynomial.
- (b) Find eigenvalues.
- (c) Find one eigenvector for each eigenvalue.

Q9. Novel

The region bounded by $y = x$ and $y = x^2$, $0 \leq x \leq 1$, is rotated about the y -axis.

- (a) Set up a shell integral.
- (b) Evaluate the volume.
- (c) Explain why washers would require rewriting in terms of y .

Q10. Pitfall

Let v_1, v_2 be eigenvectors with eigenvalues 2, -3 , and $x = v_1 + 2v_2$.

- (a) Compute Ax .
- (b) Compute A^2x .
- (c) Explain the common mistake of squaring coefficients.



W12 Applications of Integration and Eigenvectors

Q1.

Curve length for $y = f(x)$ on $[a, b]$ is

- (A) $\int_a^b f dx$
- (B) $\pi \int f^2 dx$
- (C) $\int_a^b |f'| dx$ always
- (D) $\int_a^b \sqrt{1 + (f')^2} dx$

Q2. Formula

If $A = PDP^{-1}$, then $A^5 =$

- (A) PD^5P^{-1}
- (B) $P^5D^5P^{-5}$
- (C) D^5
- (D) $P^{-1}D^5P$

Q3.

A 3×3 matrix with eigenvalues 1, 2, 3 is

- (A) not diagonalizable
- (B) diagonalizable
- (C) singular
- (D) rank 1

Q4.

The average value of f on $[a, b]$ is

- (A) $\int_a^b f$
- (B) $f(b) - f(a)$
- (C) $\frac{1}{b-a} \int_a^b f$
- (D) $f'(c)$

Q5. Pitfall

Geometric multiplicity is

- (A) root multiplicity
- (B) determinant
- (C) trace
- (D) dimension of eigenspace

Q6.

Volume by shells around the y -axis uses shell radius

- (A) x
- (B) y
- (C) $f'(x)$
- (D) λ

Q7.

If $P^{-1}AP = D$, then $P^{-1}A^3P =$

- (A) $3D$
- (B) D^3
- (C) PD^3P^{-1}
- (D) A^3

**Q8. Strategy**

Find the volume when the region under $y = \sqrt{x}$, $0 \leq x \leq 4$, is rotated about the x -axis.

- (a) Set up washers.
- (b) Evaluate.
- (c) State the radius.

Q9. Novel

Let A have eigenvectors v_1, v_2, v_3 forming a basis with eigenvalues $1, -1, 2$.

- (a) If $x = 2v_1 - v_2 + 3v_3$, compute A^4x .
- (b) Explain why the basis condition matters.
- (c) State the corresponding diagonal matrix D .

Q10. Pitfall

A 2×2 matrix has characteristic polynomial $(\lambda - 4)^2$ and eigenspace dimension 1.

- (a) Is it diagonalizable?
- (b) What are algebraic and geometric multiplicities?
- (c) Give a model matrix with this behaviour.



W13 Applications and Diagonalisation

Q1.

If $A = PDP^{-1}$, then columns of P are

- (A) eigenvectors of A
- (B) rows of A
- (C) standard basis always
- (D) determinants

Q2. Formula

For $D = \text{diag}(2, -1)$, $D^{10} =$

- (A) $\text{diag}(20, -10)$
- (B) $\text{diag}(1024, 1)$
- (C) $\text{diag}(2, 1)$
- (D) $\text{diag}(100, -1)$

Q3.

If a 4×4 matrix has 4 distinct eigenvalues, it is

- (A) singular
- (B) orthogonal
- (C) diagonalizable
- (D) not invertible

Q4.

A non-diagonalizable 2×2 example is

- (A) $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$
- (B) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
- (C) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
- (D) $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$

Q5. Pitfall

Diagonalisation is useful for

- (A) computing high powers A^n
- (B) making all matrices commute
- (C) changing determinants to zero
- (D) avoiding eigenvectors

Q6.

If $AP = PD$ and P is invertible, then

- (A) $PAP^{-1} = D$
- (B) $P^{-1}AP = D$
- (C) $A = D$
- (D) $P = D$

Q7.

Application of diagonalisation to recurrences usually computes

- (A) $\int Adx$
- (B) $\det x_0$
- (C) $x_n = A^n x_0$



(D) $A + x_0$

Q8. Strategy

Let $A = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}$.

- (a) Find A^n .
- (b) Compute $A^5(1, 1)^T$.
- (c) Explain why this is already diagonalised.

Q9. Novel

Let $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

- (a) Find eigenvalues.
- (b) Find one eigenvector for each.
- (c) Construct P, D with $A = PDP^{-1}$.

Q10. Pitfall

A student has correct eigenvalues but puts eigenvectors in a different order from D .

- (a) Explain why the order must match.
- (b) Describe how to verify using $AP = PD$.
- (c) State one consequence for computing A^n .



Answer Key

W1 Q1. A.

W1 Q2. B.

W1 Q3. C.

W1 Q4. D.

W1 Q5. A.

W1 Q6. B.

W1 Q7. C.

W1 Q8. $z = 2\sqrt{2}e^{-i\pi/4}$; $z^4 = (2\sqrt{2})^4 e^{-i\pi} = 64(-1) = -64$; $\pi/4$ is quadrant I, not IV.

W1 Q9. $16i = 16e^{i\pi/2}$, so $w_k = 2e^{i(\pi/8+k\pi/2)}$, $k = 0, 1, 2, 3$. Arguments: $\pi/8, 5\pi/8, 9\pi/8, 13\pi/8$.

W1 Q10. Example $A = \{1\}$, $B = \{2\}$ in $U = \{1, 2, 3\}$: LHS $\{3\}$, RHS $\{1, 2, 3\}$. Correct: $(A \cup B)^c = A^c \cap B^c$. Logically, $x \notin A \cup B$ means $x \notin A$ and $x \notin B$, hence intersection, not union.

W2 Q1. B.

W2 Q2. C.

W2 Q3. D.

W2 Q4. A.

W2 Q5. B.

W2 Q6. C.

W2 Q7. D.

W2 Q8. Domain $(-1/3, 0) \cup (0, \infty)$. Limit $3 - 1 = 2$. Set $F(0) = 2$.

W2 Q9. If equal square roots, square both sides to get x equal; every $y \geq 0$ is hit by $x = y^2 - 1$; $h^{-1}(y) = y^2 - 1$, $y \geq 0$.

W2 Q10. $2e^{2\pi i k/3}$, $k = 0, 1, 2$. Complex arguments can differ by 2π before division by 3.

W3 Q1. C.

W3 Q2. D.

W3 Q3. A.

W3 Q4. B.

W3 Q5. C.

W3 Q6. D.

W3 Q7. A.

W3 Q8. $\sin 2x = 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5 + \dots$, so $L = 4/15$.

W3 Q9. Dot = 5. Cross = $(-1, 5, 3)$. Area = $\sqrt{35}$.

W3 Q10. Need cubic terms: $\sin 3x - 3 \sin x = (3x - \frac{27x^3}{6}) - 3(x - \frac{x^3}{6}) + \dots = -4x^3 + \dots$; limit -4 .

W4 Q1. D.

W4 Q2. A.

W4 Q3. B.

W4 Q4. C.

W4 Q5. D.

W4 Q6. B.

W4 Q7. A.

W4 Q8. $g' = 1 - 4/x^2$, critical $x = 2$; sign changes $-$ to $+$, local/global min; $g'' = 8/x^3 > 0$, concave up.

W4 Q9. Parallel iff $\mathbf{v} \cdot \mathbf{n} = 0$. Perpendicular iff $\mathbf{v}$ is a scalar multiple of $\mathbf{n}$. Example $(1, 2, 0)$.

W4 Q10. x^3 changes increasing through 0 with no extremum; $f''(0) = 0$ inconclusive. Example x^2 .

W5 Q1. A.

W5 Q2. B.

W5 Q3. C.

W5 Q4. D.

W5 Q5. A.

W5 Q6. B.

W5 Q7. C.

W5 Q8. Substitute: $1 + t + 2 - t + 2t = 8 \Rightarrow 3 + 2t = 8$, $t = 5/2$, point $(7/2, -1/2, 5)$. The line direction is $\mathbf{v} = (1, -1, 2)$



and the plane normal is $\mathbf{n} = (1, 1, 1)$, so $\mathbf{v} \cdot \mathbf{n} = 2$ and $\cos \theta = 2/(\sqrt{6}\sqrt{3}) = \sqrt{2}/3$. Parallel to the plane iff direction vector $\cdot \mathbf{n} = 0$.

W5 Q9. Let L be limit. $\ln L = \ln \ln(x+1)/\ln x \rightarrow 0$, so $L = 1$.

W5 Q10. Use $Q - P = (1, 2, 0)$, $R - P = (2, -1, 1)$. Cross $= (2, -1, -5)$, area $\sqrt{30}$. Dot product measures projection.

W6 Q1. B.

W6 Q2. C.

W6 Q3. D.

W6 Q4. A.

W6 Q5. B.

W6 Q6. C.

W6 Q7. D.

W6 Q8. $A = 12x - 2x^2$, $A' = 12 - 4x$, critical $x = 3$. Values 0, 18, 0, max 18.

W6 Q9. $Q - P = (2, 1, -1)$, $R - P = (1, -1, 1)$, cross $(0, -3, -3)$, so $y + z = 2$.

W6 Q10. Domain $\mathbb{R} \setminus \{1\}$. $g' = -2/(x-1)^2$, no critical points. $\forall x \neq 1$, $HA \ y = 1$, $g'' = 4/(x-1)^3 > 0$ on $(1, \infty)$.

W7 Q1. C.

W7 Q2. D.

W7 Q3. A.

W7 Q4. B.

W7 Q5. C.

W7 Q6. D.

W7 Q7. A.

W7 Q8. Elimination gives $x = 1, y = 2, z = 3$.

W7 Q9. Determinant $a - 2$. At $a = 2$, equations are dependent and consistent, so infinitely many solutions.

W7 Q10. $1 = (-3) \det A$, so $\det A = -1/3$. Row replacement has factor 1.

W8 Q1. D.

W8 Q2. A.

W8 Q3. B.

W8 Q4. C.

W8 Q5. D.

W8 Q6. A.

W8 Q7. B.

W8 Q8. $1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3$. $f^{(4)} < 0$ on $0 < x < 1$, so remainder negative; polynomial overestimates.

W8 Q9. $AB = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}$, $BA = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$, not equal. $\det(AB) = \det A \det B = (-2)(-1) = 2$.

W8 Q10. Coefficient should be $1/2$, not $1/24$: substitute $u = -x^2$ into $e^u = 1 + u + u^2/2 + \dots$.

W9 Q1. A.

W9 Q2. B.

W9 Q3. C.

W9 Q4. D.

W9 Q5. A.

W9 Q6. B.

W9 Q7. C.

W9 Q8. $\int_0^1 (1+x)^2 dx = [(1+x)^3/3]_0^1 = 7/3$.

W9 Q9. $F' = e^x \sqrt{1+e^{2x}} - \sqrt{1+x^2}$. $F'(0) = \sqrt{2} - 1$. Lower limit subtracts $f(x)x'$.

W9 Q10. The expression equals $\sum k^2/n^2 \sim n/3$, diverges. A true sum needs $(k/n)^2(1/n) = k^2/n^3$.

W10 Q1. B.

W10 Q2. C.

W10 Q3. D.

W10 Q4. A.

W10 Q5. B.



W10 Q6. C.

W10 Q7. D.

W10 Q8. Limits $0 \rightarrow 4$, $I = \int_0^4 \frac{2u}{(u+1)(u+3)} du = \int_0^4 \left(-\frac{1}{u+1} + \frac{3}{u+3} \right) du = 3 \ln 7 - \ln 5 - 3 \ln 3$.

W10 Q9. $\text{Det} = 1(3) - 2(-2) = 7$, invertible; unique solution for every b .

W10 Q10. Singularity at 0 requires a limit; $\lim_{a \rightarrow 0^+} \int_a^1 x^{-1} dx = \infty$, diverges. $\int_0^1 x^{-1/2} dx = 2$.

W11 Q1. C.

W11 Q2. D.

W11 Q3. A.

W11 Q4. B.

W11 Q5. C.

W11 Q6. D.

W11 Q7. A.

W11 Q8. $\lambda^2 + \lambda - 6 = (\lambda - 2)(\lambda + 3)$. For $\lambda = 2$, $(3, 1)^T$; for $\lambda = -3$, $(-2, 1)^T$.

W11 Q9. $V = 2\pi \int_0^1 x(x - x^2) dx = 2\pi(1/3 - 1/4) = \pi/6$.

W11 Q10. $Ax = 2v_1 - 6v_2$, $A^2x = 4v_1 + 18v_2$. Coefficients stay outside; eigenvalues are powered.

W12 Q1. D.

W12 Q2. A.

W12 Q3. B.

W12 Q4. C.

W12 Q5. D.

W12 Q6. A.

W12 Q7. B.

W12 Q8. $V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \int_0^4 x dx = 8\pi$.

W12 Q9. $A^4x = 2v_1 - v_2 + 48v_3$. Basis lets every vector decompose uniquely. $D = \text{diag}(1, -1, 2)$.

W12 Q10. Not diagonalizable. Algebraic multiplicity 2, geometric 1. Example $\begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix}$.

W13 Q1. A.

W13 Q2. B.

W13 Q3. C.

W13 Q4. D.

W13 Q5. A.

W13 Q6. B.

W13 Q7. C.

W13 Q8. $A^n = \text{diag}(3^n, (-2)^n)$. $A^5(1, 1)^T = (243, -32)^T$. Diagonal entries act independently.

W13 Q9. Char poly $\lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2)$. Eigenvectors: for 5, $(1, 1)^T$; for 2, $(1, -2)^T$. $P = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}$, $D = \text{diag}(5, 2)$.

W13 Q10. Column i of P must match diagonal entry D_{ii} . Verify each column equation $Av_i = \lambda_i v_i$. Wrong order gives wrong powers.